

TIANYI CAI

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SUMMARY

- Profile** · Applied scientist with Biostatistics PhD and 9 years of experience combining causal inference, experimentation, machine learning, and scalable engineering to drive strategic decision-making.
- Skills** · Causal inference: doubly robust estimators, quasi-experimental methods, heterogeneous treatment effects.
· Machine learning: tabular/panel/time series data, temporal feature engineering, model selection and validation.
· Experimentation: study design, power and minimum detectable effect, bandit algorithms.
· Engineering: CI/CD pipelines, distributed computing, model deployment and monitoring.
· Leadership: cross-functional collaboration, thought leadership, stakeholder communication and management.
- Toolkit** · Python, SQL, R, Anyscale/Ray, MLflow, Airflow, Snowflake, Databricks, Claude Code.

EXPERIENCE

Staff Applied Scientist, Consumer Trading **Coinbase**, New York, NY 01/2025–Present

- Architected agentic workflows for causal inference using custom skills/MCPs; scaled incrementality analyses from weeks to hours, shaping product go/no-go decisions and incentive sizing (Coinbase One Card, new product/asset cannibalization).
- Deployed end-to-end machine learning models predicting customer LTV (long-term experimentation metric), trading volume, and churn (high-net-worth targeting); outputs inform strategy across marketing, sales, and CX.
- Engineered unified temporal data pipelines powering machine learning and causal inference models across the broader consumer organization, leveraging daily data snapshots to ensure point-in-time correctness and prevent data leakage.
- Operationalized a time series anomaly detection service that flags weeks where consumer trading metrics fall outside their prediction intervals, then drills down to the user level to produce one-pagers for sales teams.

Senior Data Scientist, Planner Evaluation **Waymo**, New York, NY 11/2022–01/2025

- Halved the family-wise error rate in onboard software evaluation by applying confidence interval adjustments.
- Pioneered the use of statistical power and minimum detectable effect as evaluation metrics throughout the Planner Evaluation team by authoring the API and supporting documentation.
- Provided statistical mentorship and workshops for teams across Waymo on topics such as resampling methods, Type I/II error, and quantifying uncertainty.

Senior Research Data Scientist, Quantitative Engineering **Meta**, New York, NY 02/2021–11/2022

- Doubled user engagement with infrastructure tools by replacing unreliable front-end North Star metrics with robust back-end metrics and identifying their causal drivers, in collaboration with engineering, product, design, and UX research.
- Segmented developer workflows by fine-tuning a pre-trained transformer model to extract high-dimensional vector embeddings for computational notebooks; directly informed the engineering roadmap for user-level custom features.
- Audited the experimentation strategy for measuring p99 latency in query services; designed a formal statistical framework incorporating variance reduction and alternative metrics.

Applied Scientist II, Global Intelligence **Uber**, San Francisco, CA 01/2020–02/2021

- Quantified uncertainty in competitive intelligence metrics, providing statistically rigorous signals that underpinned executive decisions on global market exits and consolidations.
- Ensembled machine learning models using internal and third-party data to identify rare user groups for multimillion-dollar experiments; improved the legacy model by doubling average precision.

Senior Data Scientist, Data and Research **BitSight**, Boston, MA 10/2017–01/2020

- Led all data science projects for third-party cybersecurity risk management, including BitSight's inaugural machine learning product (patent), multiple analytical tools, and observational studies to evaluate internal processes.
- Supervised a Master's-level intern working on validation of external financial data and sales/marketing analytics.

EDUCATION

PhD, Biostatistics **Harvard University** 2017

BMath, Statistics and Actuarial Science **University of Waterloo** 2012